Predicting Reaction Time with Vital Signs

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INTRODUCTION

- Our data was produced by the Korea Sports Promotion Foundation or KSPO
- The data is itemized measurements of the National Physical Fitness Measurement Data managed by the Seoul Olympic Commemorative National Sports Promotion Corporation for athletes of all levels and ages
- The dataset included 1083 observations with desired variables of age, sex, height, weight, body fat percentage, systolic/diastolic blood pressure, resting heartrate, and reaction time
- Reaction time has been used since the late 19th century to be a measurement of "processing speed and measures of fluid intelligence."
- Some studies predict that "In high-income populations, severe obesity is expected to double its prevalence from 10 to 20% between 2020 and 2035"
- We also found the beginnings of research linking obesity to poor memory deficits and brain shrinkage
- The purpose of the Multiple Linear Regression Model (MLR) was to investigate the relationship between general health markers and reaction time while attempting to account for factors like age and sex
- The purpose of the Chi-Square Test of Independence was to investigate to examine the relationship between body fat percentage and reaction time

METHODS

Data Cleaning

- Observations missing any desired variable were excluded
- All variables were trimmed, removing the top 1% and bottom 1% of values

Descriptive Analysis

- Descriptive statistics including min, max, mean, median, standard deviation, skewness and kurtosis were produced and reviewed for quantitative variables
- Plots generated included histogram, quantile-quantile, boxplot
- Frequency tables generated for categorical variables

Inferential Analyses

- A multiple linear regression analysis was conducted using reaction time as the dependent variable with an of alpha = 0.05
- Independent variables included age, sex, height, weight, body fat %, diastolic bp, systolic bp, and resting heart rate
- Assumptions of MLR were assessed including correct model specification (residual scatterplot), normality of residuals (residual scatterplot, histogram, qq plot, skewness, kurtosis), equal variance of residuals (residual scatterplot), independence of residuals (study design), and no multicollinearity (variance inflation factor)
- A chi-square test of independence was conducted to examine the relationship between body fat percentage and reaction time with an alpha of 0.05
- Assumptions of the chi-square test were assessed including variables are categorical, assumed random sampling, independent categories, and expected cell counts of at least 5

RESULTS

Table 1: Descriptive Statistics

Variable	Type	Minimum	Maximum	Mean	Median	Std Dev
Age	Quantitative	19.0	64.0	37.6	30.0	15.8
Sex	Categorical	-	-	-	-	-
Height (cm)	Quantitative	143.2	184.4	165.9	165.5	7.5
Weight (kg)	Quantitative	41.2	100.5	62.7	60.2	11.6
Body Fat %	Quantitative	9.8	45.5	27.3	27.1	6.9
Diastolic BP (mmHg)	Quantitative	49.0	99.0	75.0	75.0	10.9
Systolic BP (mmHg)	Quantitative	96.0	159.0	126.2	124.0	14.8
Resting Heart Rate (bpm)	Quantitative	60.0	120.0	86.5	87.0	11.7
Reaction Time (sec)	Quantitative	0.3	0.7	0.4	0.4	0.1

Figure 1: Histogram of Reaction Time

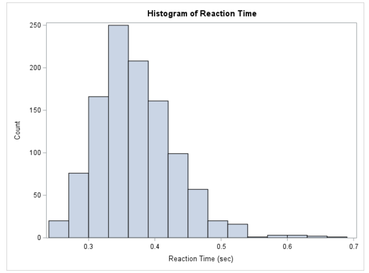


Figure 2: Residual Scatterplot, Residual QQ Plot, Residual Histogram (Before Data Trim)

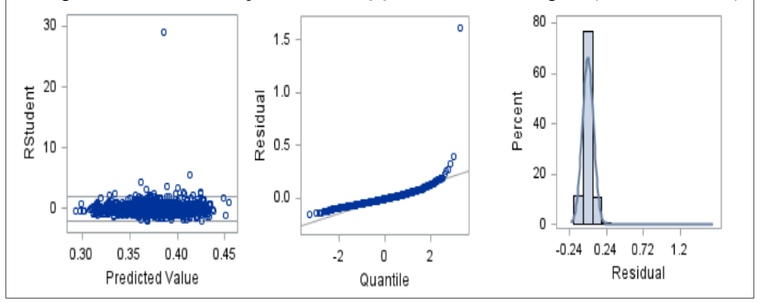


Figure 3: Residual Scatterplot, Residual QQ Plot, Residual Histogram (After Data Trim)

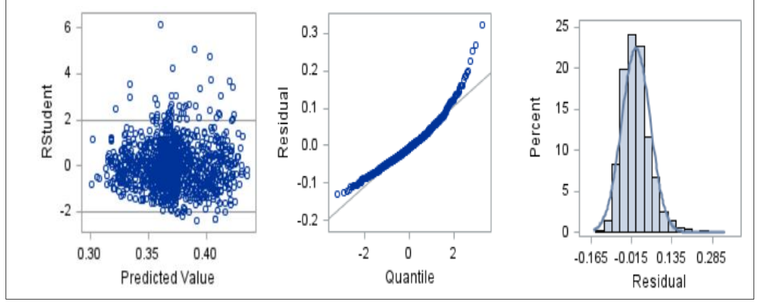


Table 2: Residual Descriptive Statistics – Univariate Output

Moments			
N	1083	Sum Weights	1083
Mean	-0.000042	Sum Observations	-0.0454683
Std Deviation	1.0005415	Variance	1.0010833
Skewness	0.98964719	Kurtosis	2.67018568
Uncorrected SS	1083.17213	Corrected SS	1083.17213
Coeff Variation	-2383171.2	Std Error Mean	0.03040331

Table 3: Residual Descriptive Statistics – Univariate Output

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.956859	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.060805	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.161165	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	7.305867	Pr > A-Sq	<0.0050

Table 4: Multiple Linear Regression Output – Parameter Estimates

Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	0.39229	0.06836	5.74	<.0001	.	0
Age	1	0.00090559	0.00013297	6.81	<.0001	0.59507	1.68047
Sex	1	-0.03027	0.00693	-4.36	<.0001	0.24558	4.07197
Height (cm)	1	-0.00027435	0.00040794	-0.67	0.5014	0.28270	3.53730
Weight (kg)	1	0.00048105	0.00029093	1.65	0.0985	0.23217	4.30714
Body Fat %	1	0.00116	0.00039436	2.94	0.0034	0.35286	2.83401
Diastolic Blood Pressure (mmHg)	1	-0.00033601	0.00020174	-1.67	0.0961	0.54195	1.84519
Systolic Blood Pressure (mmHg)	1	-0.00006534	0.00016255	-0.40	0.6878	0.45730	2.18675
Resting Heart Rate (bpm)	1	-0.00031630	0.00014633	-2.16	0.0309	0.90088	1.11003

Table 5: Multiple Linear Regression Output

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	0.70097	0.08762	30.66	<.0001
Error	1074	3.06924	0.00286		
Corrected Total	1082	3.77021			

Root MSE	0.05346	R-Square	0.1859
Dependent Mean	0.37164	Adj R-Sq	0.1799
Coeff Var	14.38452		

Table 6: Contingency Table of Body Fat Percentage and Reaction Time

Frequency Expected Cell Chi-Square Percent Row Pct Col Pct	Table of bf_cat by rt_cat		
	rt_cat		
bf_cat	Faster RT	Slower RT	Total
Avera	268 212.64 14.414 24.75 63.66 48.99	153 208.36 14.71 14.13 36.34 28.54	421 38.87
Obese	279 334.36 9.1666 25.76 42.15 51.01	383 327.64 9.3547 35.36 57.85 71.46	662 61.13
Total	547 50.51	536 49.49	1083 100.00

Table 7: Chi-Square Statistics

Statistic	DF	Value	Prob
Chi-Square	1	47.6449	<.0001
Likelihood Ratio Chi-Square	1	48.1106	<.0001
Continuity Adj. Chi-Square	1	46.7881	<.0001
Mantel-Haenszel Chi-Square	1	47.6009	<.0001
Phi Coefficient		0.2097	
Contingency Coefficient		0.2053	
Cramer's V		0.2097	

MLR Assumptions and Interpretation

- Correct model specification was supported by residual scatterplot (Figure 3) showing no curves, groups, or clusters
- Normality of residuals was supported by residual scatterplot, histogram, QQ plot (Figure 3) as well as skewness and kurtosis (Table 2) however Tests for normality did not support normality (Table 3)
- Equal Variance of Residuals supported by residual scatterplot (Figure 3) showing approximately unbiased and homoscedastic and no cone pattern evident
- Independence of Residuals assumed supported based on information on dataset
- No Multicollinearity supported by no VIF values above 5 (Table 4)
- Approximately 18.6% of the variation in reaction time is explained by the independent variables age, sex, height, weight, body fat %, diastolic bp, systolic bp, resting heart rate, f(8, 1074)=30.66, p<.001, R^2=0.1859 (Table 5)

Chi-Square Assumptions and Interpretation

- Assumptions were supported including variables were categorical, random sampling, independent categories, and expected values at least 5 (Table 6)
- There is a significant association between body fat percentage and reaction time, $\chi^2(1, n=1083) = 47.65, p < .001$). Individuals with lower body fat percentages tend to have faster reaction times compared to those with higher body fat percentages (Table 6)

Outliers and Missing Data

- Due to observations with extreme outliers, we opted to trim the top 1% and bottom 1% from the data
- Out of 12,197 observations in the data set, only 1083 contained data for all variables of interest

DISCUSSION

Issues

Extreme outliers were present initially including data that seemed impossible. Although we could have treated this as data entry error, being as this was biological data and we do not count ourselves content experts, we opted to do the 1% top and bottom trim to the data considering our large sample size to normalize our data fairly.

Limitations

Although this data set was publicly available with a well-documented code book, we were left with some questions on how the metrics were obtained which could affect applications of the findings.

Statistical Vs. Clinical Significance

Although the MLR was statistically significant, the prediction power seemed low given approximately 18.6% of the variation in reaction time could be explained. However, our Chi-Square test of independence did find a statistically signification relationship between body fat % and reaction time. This leads us to believe that further research in this area could potentially provide clinical significance given further refined study design and included content experts. This could be important as obesity (high body fat %) has been linked with many risks and perhaps reaction time could be one more metric available to assess health or screen for conditions.